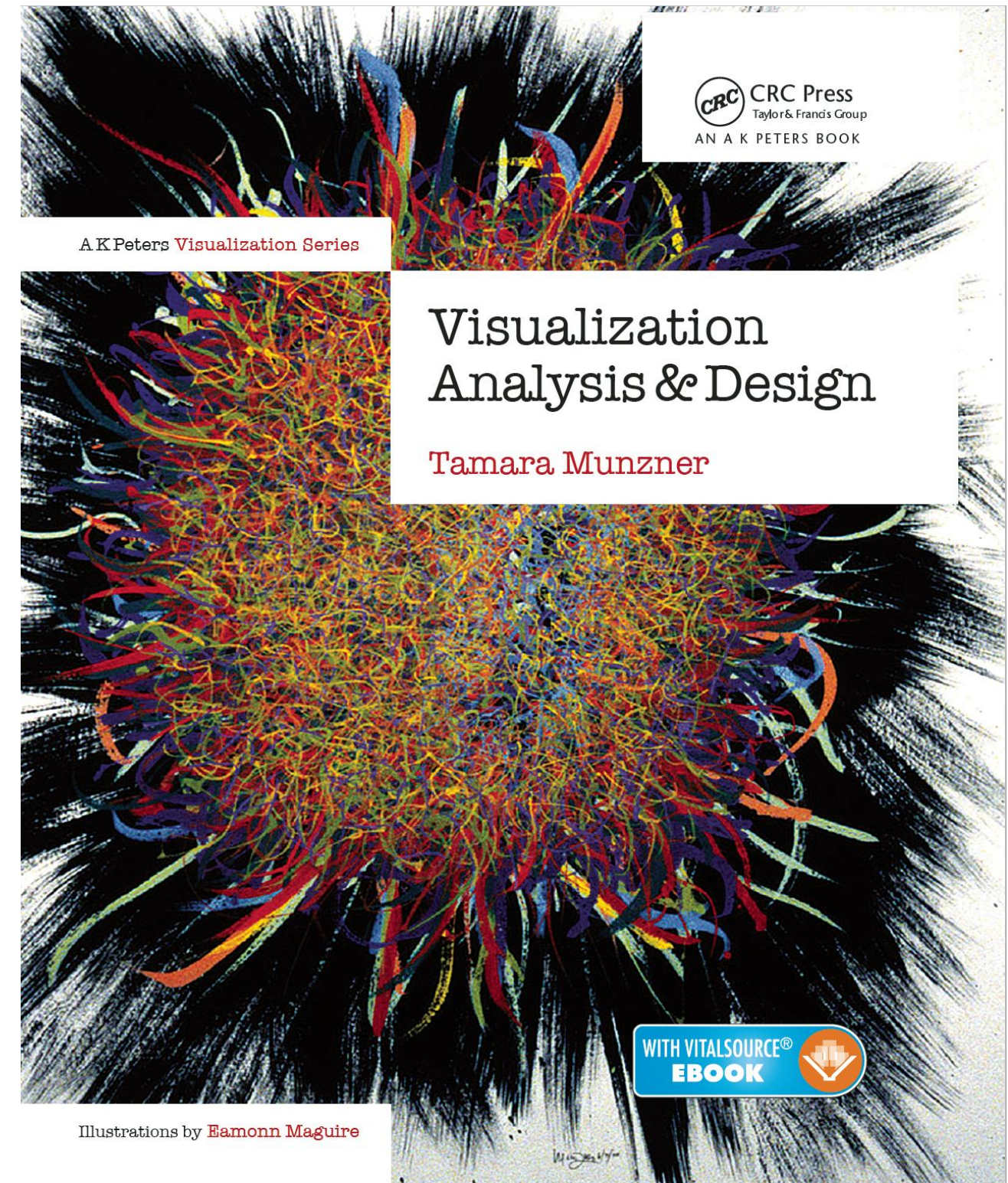


# CDS505

## Data Visualization & Visual Analytics

### Multiple Views

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- ✓ Facet
- ✓ Juxtapose and Coordinate Views
- ✓ Partition into Views
- ✓ Superimpose Layers

The verb **facet** means to split; splitting up the display, into either multiple views or layers.

1. Juxtaposing and coordinate views
2. Views side by side
3. Superimposing layers within a single view

## ➔ Juxtapose and Coordinate Multiple Side-by-Side Views

➔ Share Encoding: Same/Different

➔ *Linked Highlighting*



➔ Share Data: All/Subset/None

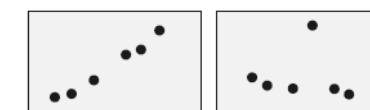


➔ Share Navigation

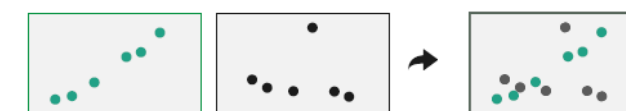


|          |           | Data                                                                                            |                                                                                                                         |                                                                                                       |
|----------|-----------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|          |           | All                                                                                             | Subset                                                                                                                  | None                                                                                                  |
| Encoding | Same      | Redundant                                                                                       |  Overview/<br>Detail               |  Small Multiples |
|          | Different |  Multiform |  Multiform,<br>Overview/<br>Detail | No Linkage                                                                                            |

## ➔ Partition into Side-by-Side Views



## ➔ Superimpose Layers



## 1. Share Encoding: Same/Different

- **Linked highlighting**
- System: **EDV (Exploratory Data Visualizer)**
- see how regions contiguous in one view are distributed within another
- powerful and pervasive interaction idiom encoding: different
  - *Multiform*

# Juxtapose and Coordinate Views

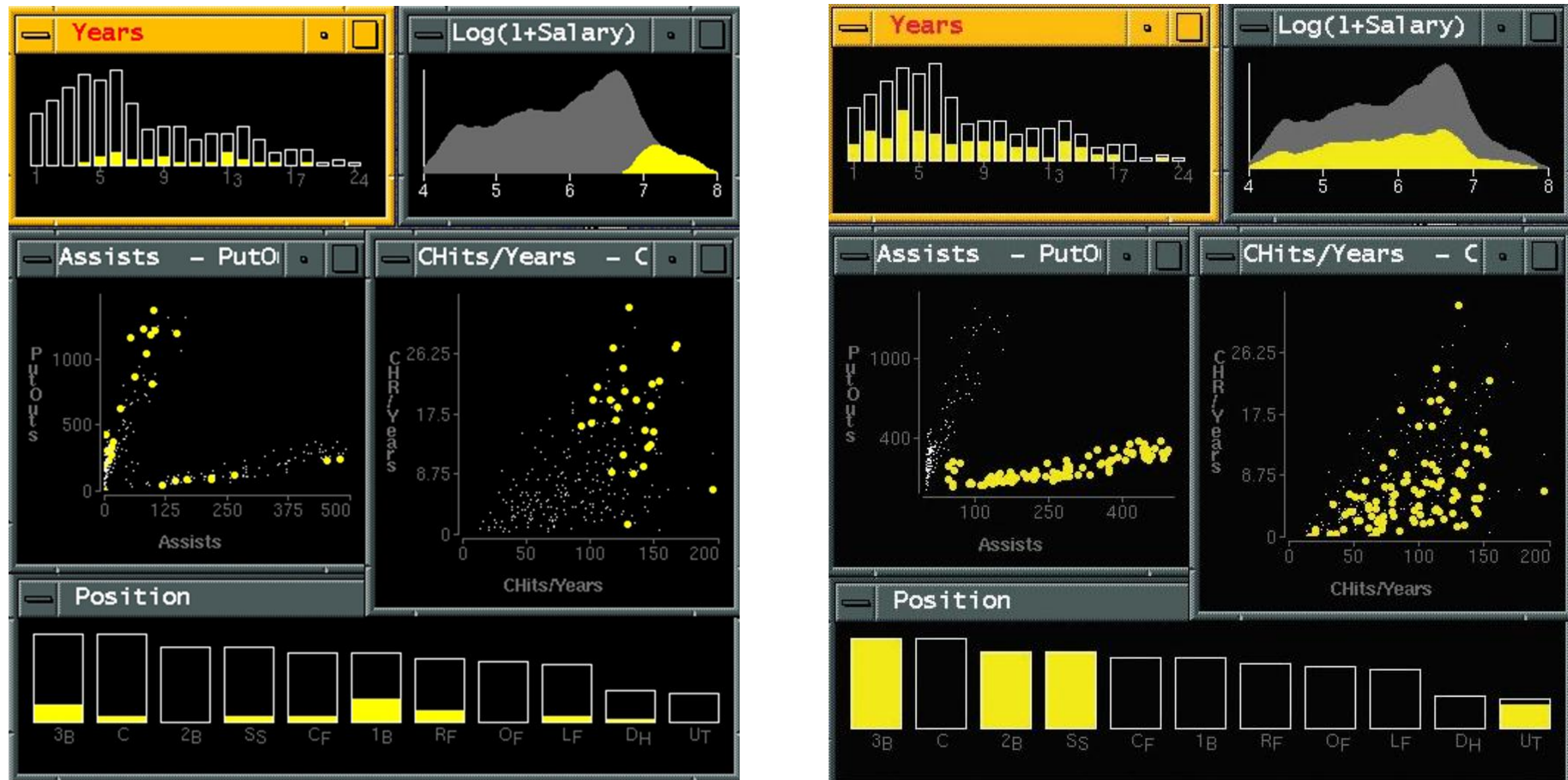


Figure 12.2: (a) Selecting the high salaries in the upper right window shows different distributions in the other views. (b) Selecting the bottom group in the *Assists-PutOuts* window shows that the clump corresponds to specific positions played. From [Wills 95].

## 2. Share Data: All, Subset, None

- **All** – shared data, both views could each show all of the data;
- **Subset - overview–detail**, one view could show a subset of what is in the other,
- **None – small multiples**, the views could show different partitions of the dataset into disjoint pieces.
- A common choice is to have only two views, one for overview and one for detail.
- When the dataset has multilevel structure at discrete scales, multiple detail views may be appropriate to show structure at these different levels (other views act as a concise visual history)

# Juxtapose and Coordinate Views

## Bird's-Eye Maps

- Navigation: shared bidirectional linking
- **Overview–detail**
- Geographic maps, the views have the same encoding and dataset; they differ in viewpoint and size.  
[<http://maps.google.com>]



- **Overview–detail** for data sharing with multiform views, where the detail view has a different visual encoding than the overview, sometimes called a **detail-on-demand view**.
- **Multiform Overview–Detail: Microarrays(next slide)**
- It features coordination between the scatterplot view in the centre and the graph view in the upper left.
- The designers carefully analysed the domain situation to generate an appropriate data and task abstraction and concluded that no single view would suffice.

# Juxtapose and Coordinate Views

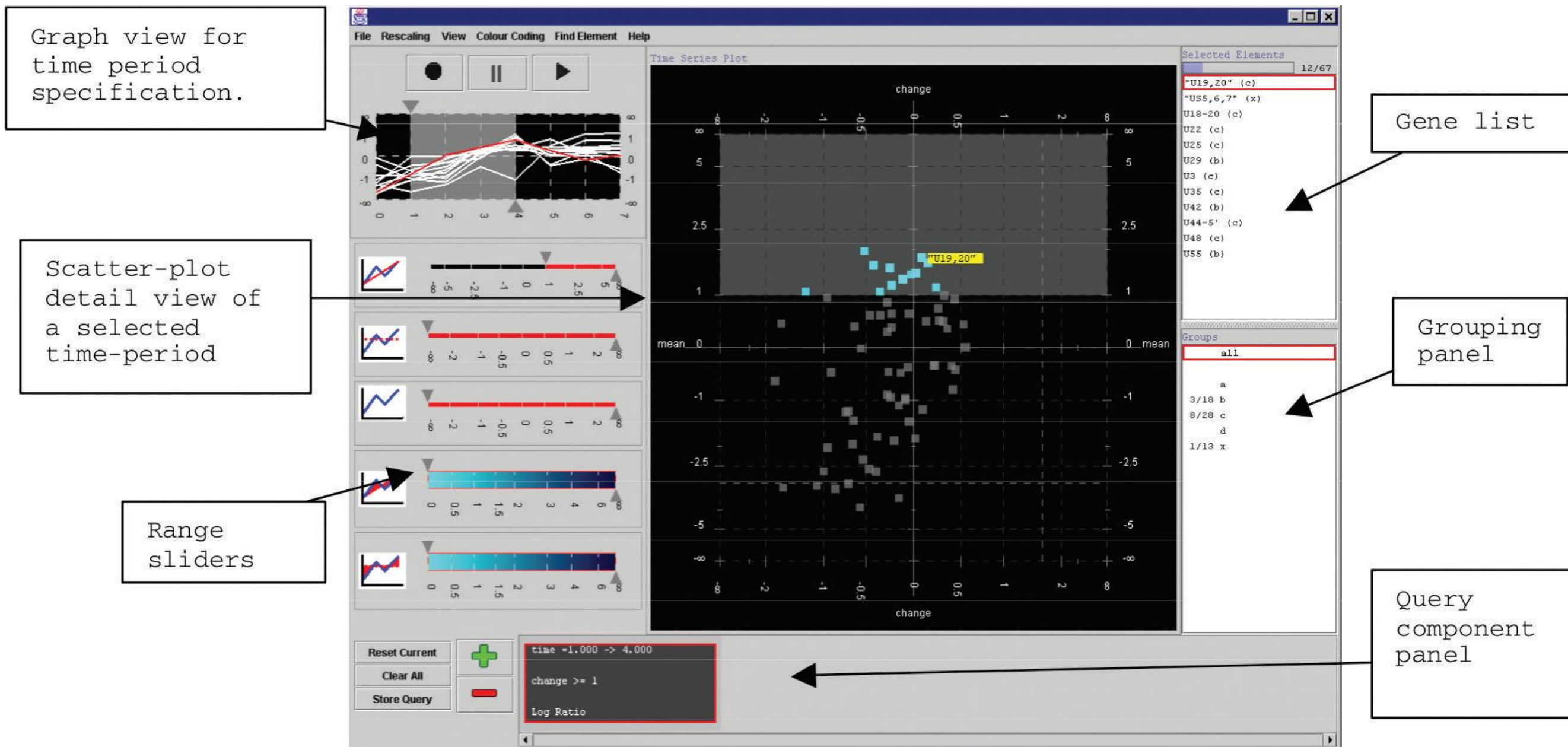


Figure 12.4: Multifunction overview–detail vis tool for microarray exploration features a central scatterplot linked with the graph view in the upper left. From [Craig and Kennedy 03].

- **Multiple views** with the same encoding and different partitions of the data between them, are often called **small multiples**.
- The shared visual encoding means that the views have a common reference frame so that comparison of spatial position between them is directly meaningful.
- Small multiples are often aligned into a list or matrix to support comparison with the highest precision. The choice of small-multiple views is in some sense the inverse of multiform views, since the encoding is identical but the data differs.



*We lead*

- From [Barsky et al. 08, Figure 2].



## 3. Share Navigation: Synchronize

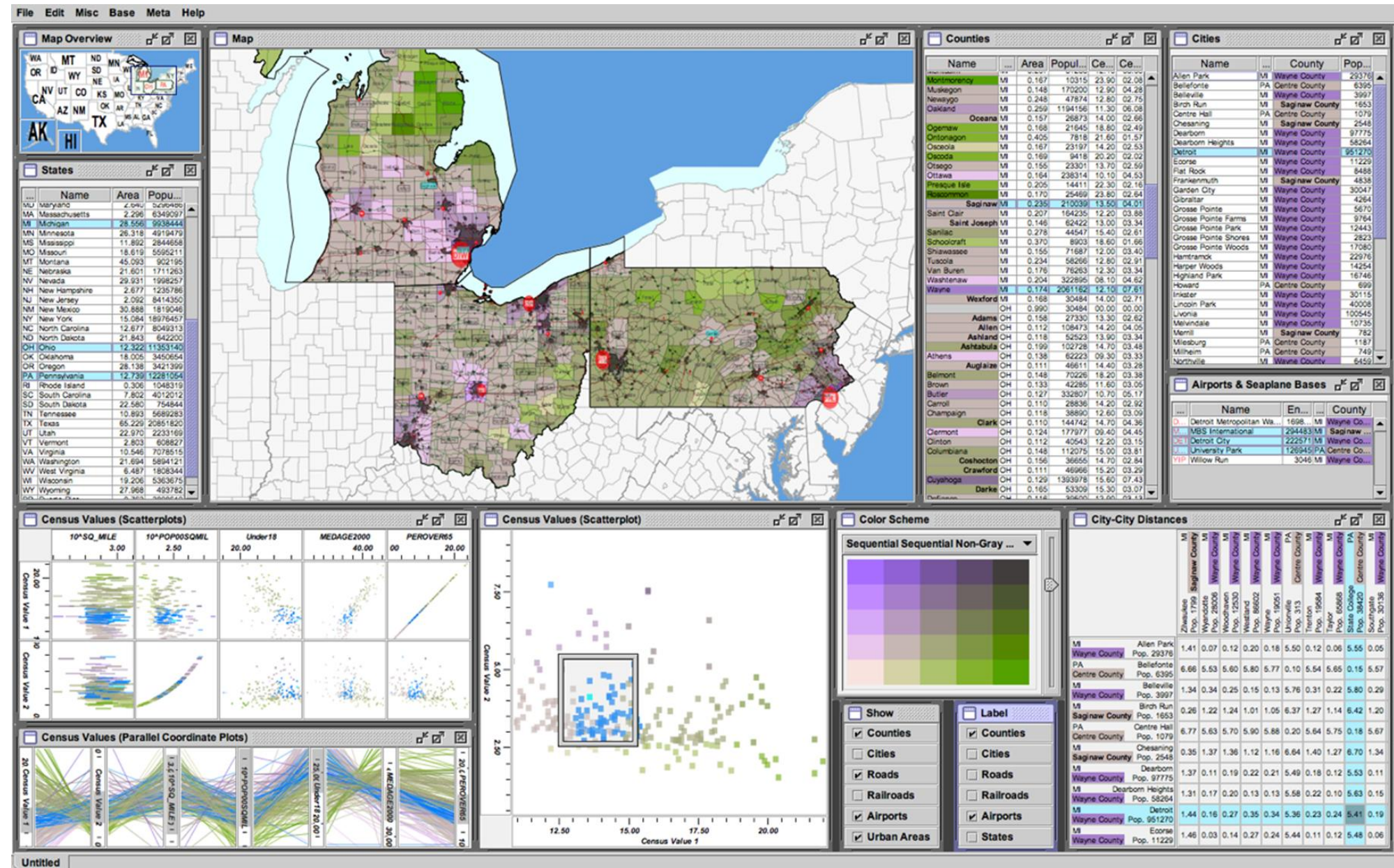
- It is another way to coordinate between views
- Linked navigation – synchronized movement between one viewpoint with the others

# Juxtapose and Coordinate Views

Figure 12.7: The Improvise toolkit [Weaver 04] - census vis that has many forms of coordination between views.

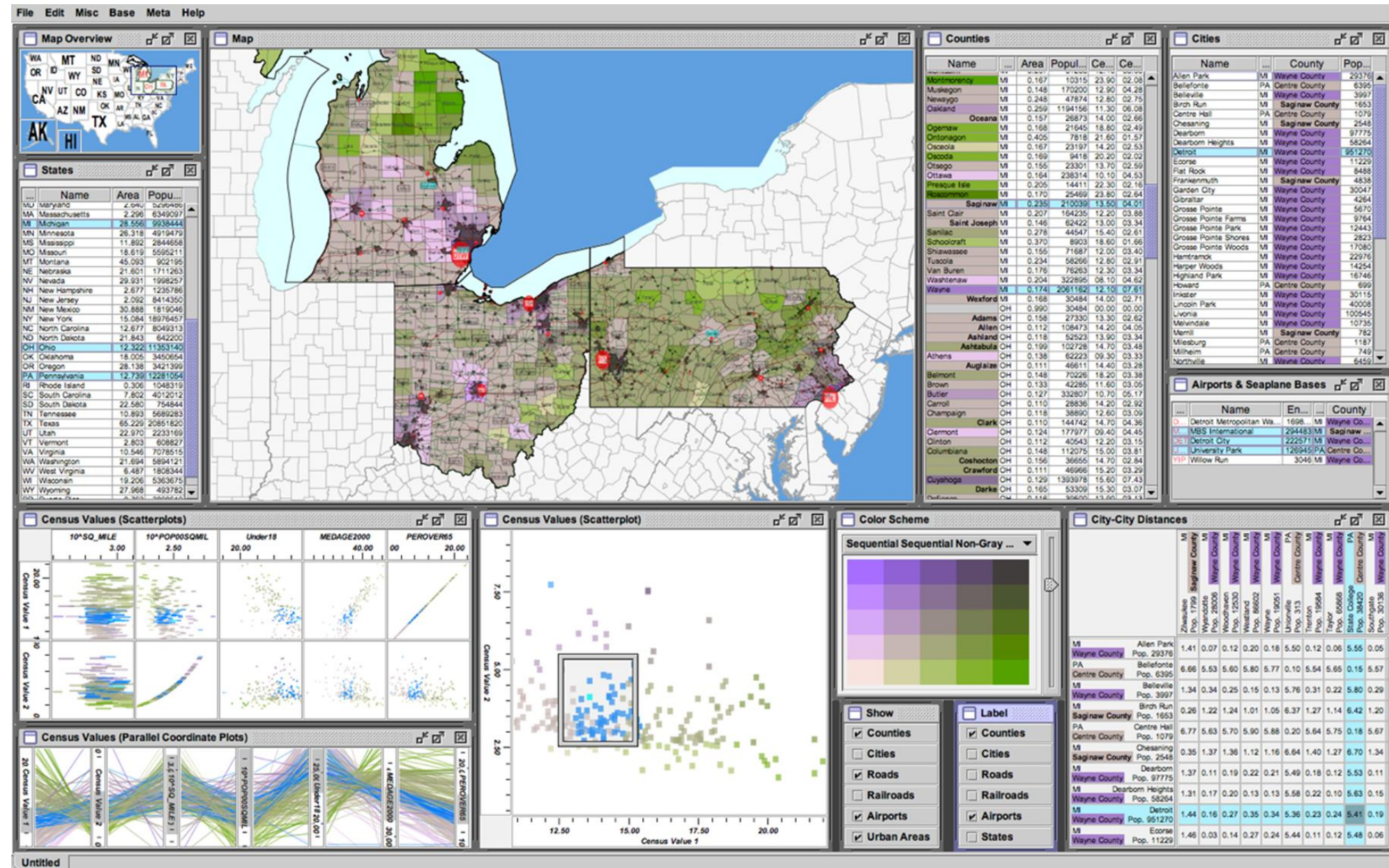
- It has many multiform views, some of which use small multiples, and some of which provide additional detail information.

<http://www.cs.ou.edu/~weaver/improvise/examples/census>



# Juxtapose and Coordinate Views

- investigate power of multiple views
  - pushing **limits** on view count, interaction complexity
  - how many is ok?
    - open research question
  - **reorderable** lists
    - easy lookup
    - useful when linked to other encodings



# Juxtapose and Coordinate Views

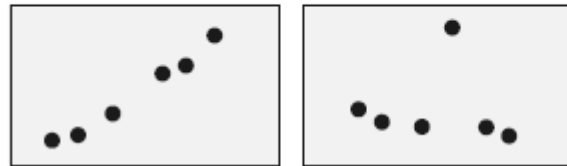
Figure 12.6: Design choices for how to coordinate between views relating to sharing encoding and data.

|          |           | Data                                                                                          |                                                                                                                        |                                                                                                     |
|----------|-----------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|          |           | All                                                                                           | Subset                                                                                                                 | None                                                                                                |
| Encoding | Same      | Redundant                                                                                     |  Overview/<br>Detail                |  Small Multiples |
|          | Different |  Multiform |  Multiform,<br>Overview/<br>Detail | No Linkage                                                                                          |

## Why juxtapose views?

- **benefits:** eyes vs. memory
  - lower cognitive load to move eyes between 2 views than remembering previous state with single changing view
- **costs:** display area, 2 views side by side each have only half the area of one view

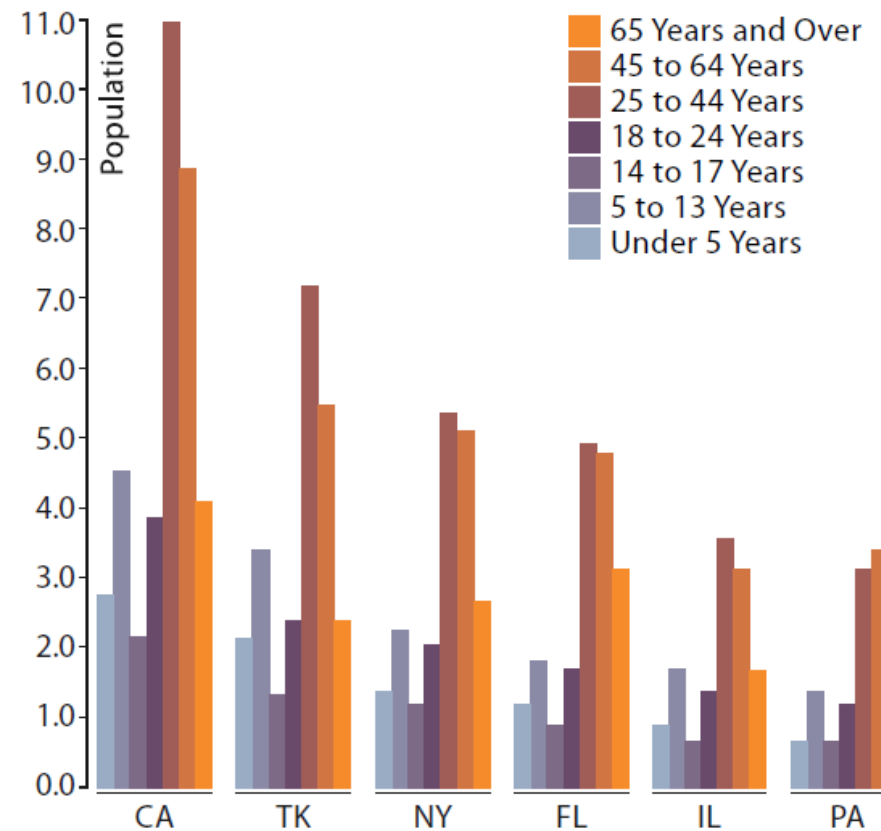
## ➔ Partition into Side-by-Side Views



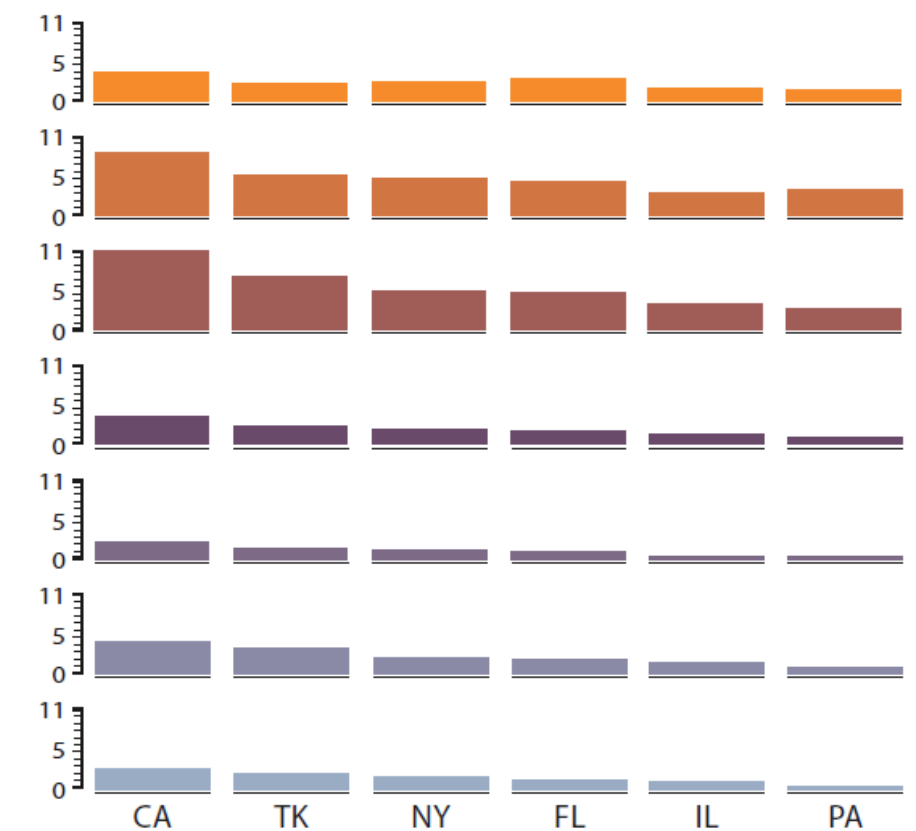
- How to divide data between views
  - split into regions by attributes
  - encodes association between items using spatial proximity
  - order of splits has major implications for what patterns are visible
- No strict dividing line
  - *view: big/detailed*
    - contiguous region in which visually encoded data is shown on the display
  - *glyph: small/iconic*
    - object with internal structure that arises from multiple marks

## List Alignment

- single bar chart with grouped bars
  - split by state key into regions
    - complex glyph within each region showing all ages
  - compare: easy within state, hard across ages
- small-multiple bar charts
  - split by age into regions
    - one chart per region
  - compare: easy within age, harder across states



(a)

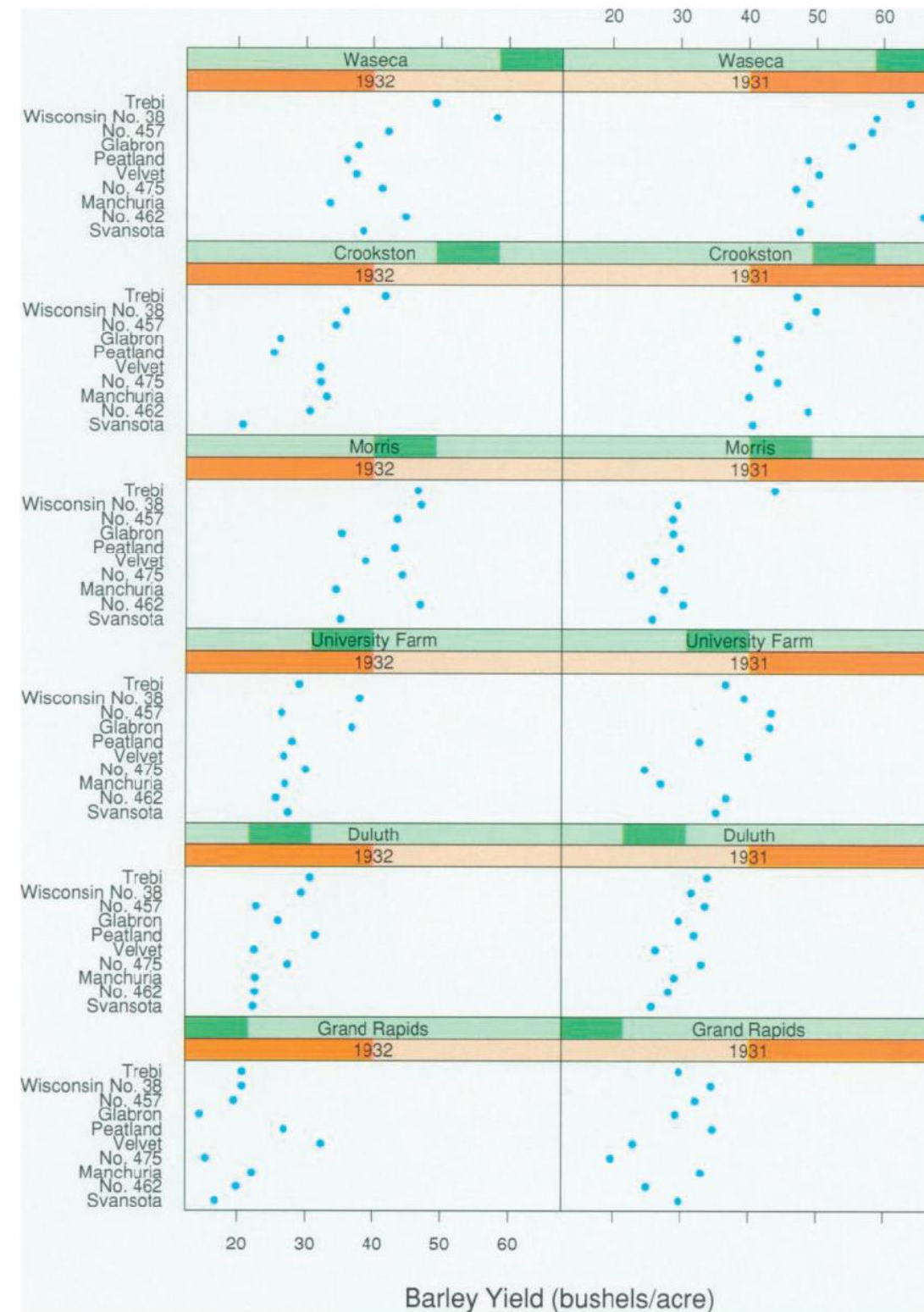


(b)

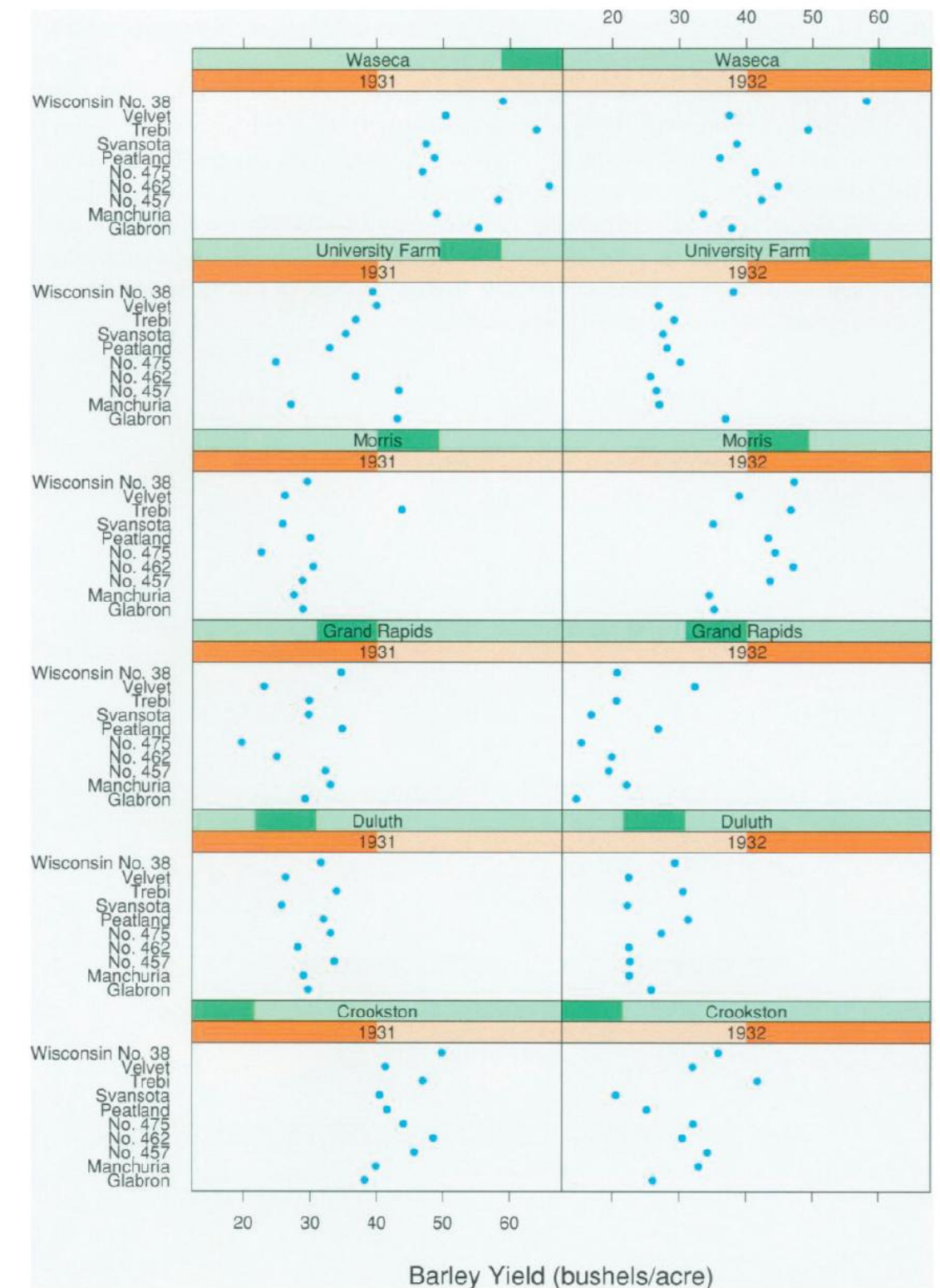
# Partition into Views

## Matrix alignment

- Partitioning a multi-attribute dataset



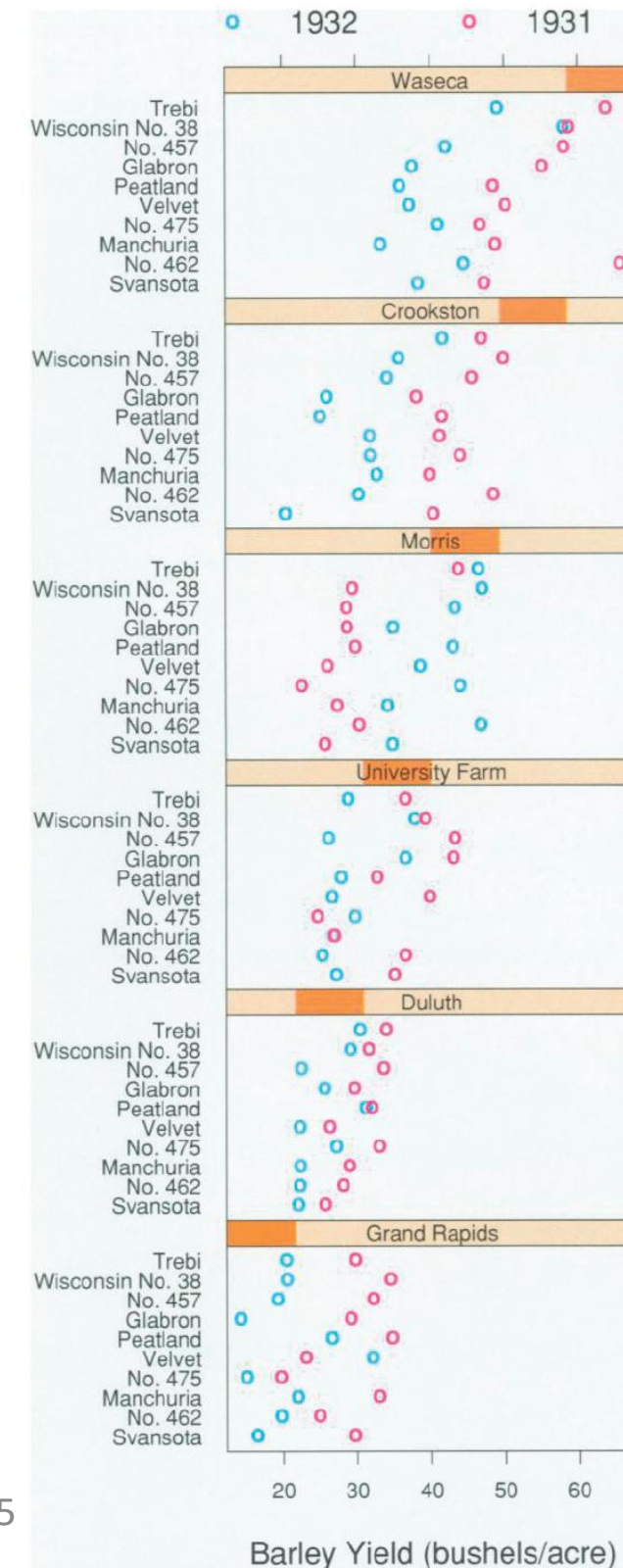
(a)



(b)

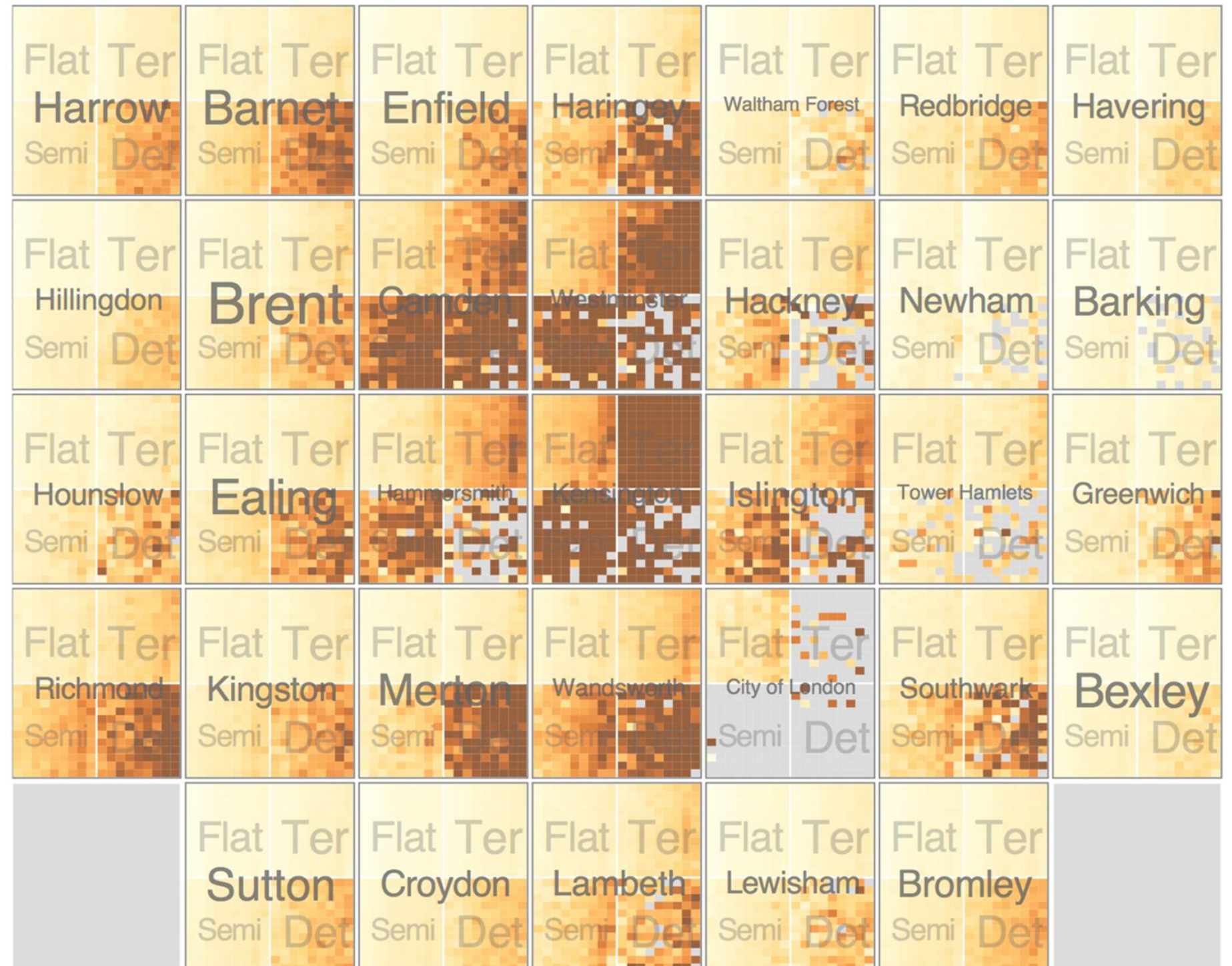
# Partition into Views

- A second Trellis plot combines the years into a single plot with year encoded by colour, showing strong evidence of an anomaly in the data. From [Becker et al. 96]



## Recursive subdivision – **HiVE** system

- split by neighbourhood
- then by type
- then time
  - years as rows
  - months as columns
- colour by price
- neighbourhood patterns
  - where it's expensive
  - where you pay much more for detached type



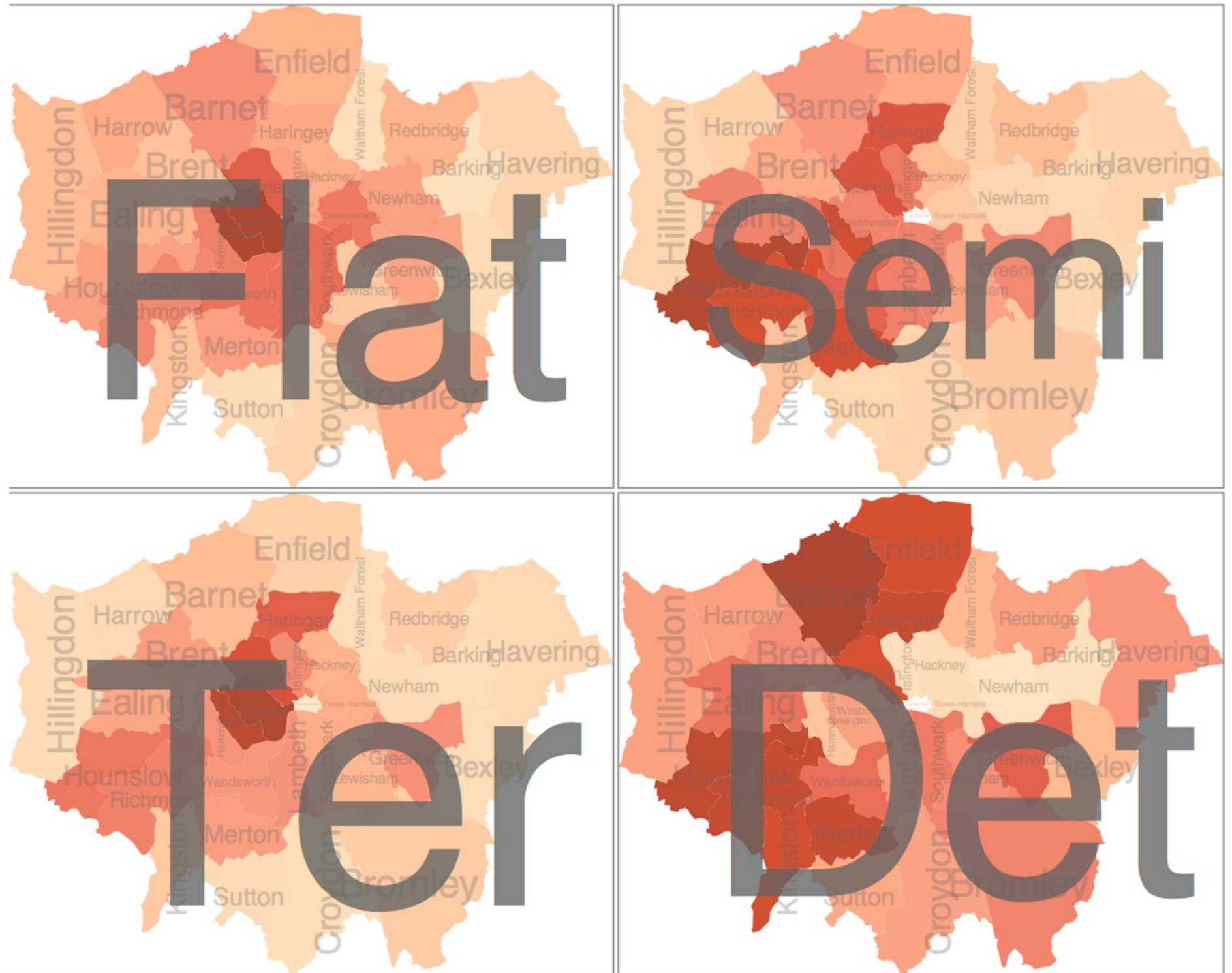
# Partition into Views

- switch order of splits
  - type then neighbourhood
- switch colour
  - by price variation
- type patterns
  - within specific type, which neighbourhoods inconsistent



# Partition into Views

- different encoding for second-level regions  
– choropleth maps

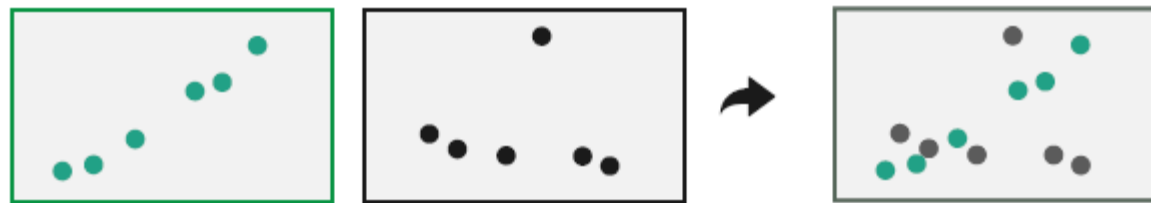


# Partition into Views

- Size regions by sale counts
  - not uniformly
- Result: treemap



## ➔ Superimpose Layers



- **Layer:** set of objects spread out over region
  - each set is visually distinguishable group
  - extent: whole view
- Design choices
  - how many layers, how to distinguish?
    - encode with different, non-overlapping channels
    - two layers achievable, three with careful design
  - small static set, or dynamic from many possible?

# Superimpose Layers

Figure 12.13: Static visual layering

- **foreground layer:** roads
  - hue, size distinguishing main from minor
  - high luminance contrast from background
- **background layer:** regions
  - desaturated colors for water, parks, land areas
- user can selectively focus attention
- “get it right in black and white”
  - check luminance contrast with greyscale view



# Superimpose Layers

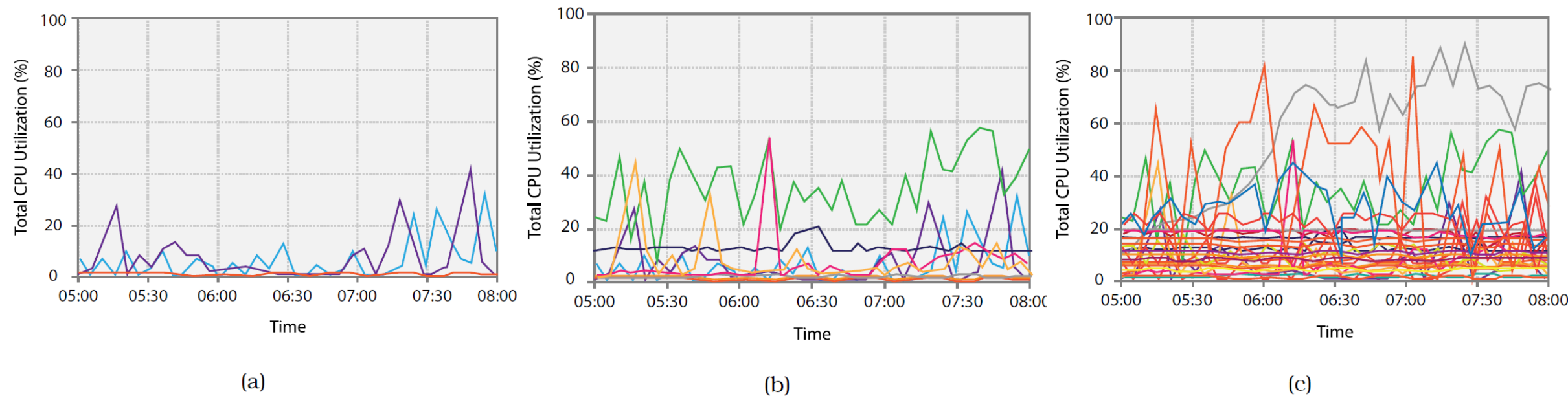


Figure 12.14: Multiple line charts can be superimposed within the same global frame.

- Few layers, but many lines
  - up to a few dozen
  - but not hundreds
- **superimpose** vs **juxtapose**: empirical study
  - superimposed for local, multiple for global
  - tasks
    - local: maximum, global: slope, discrimination
  - same screen space for all multiples vs single superimposed

# Superimpose Layers

Empirical study comparing **superimposed line charts** to juxtaposed filled-area line charts.

- (a) Superimposed line charts performed **best** for tasks carried out within a local visual span.
- (b) Juxtaposed filled area charts were **best** for global tasks, especially as the number of time series increased.

From [Javed et al. 10, Figures 1 and 2].

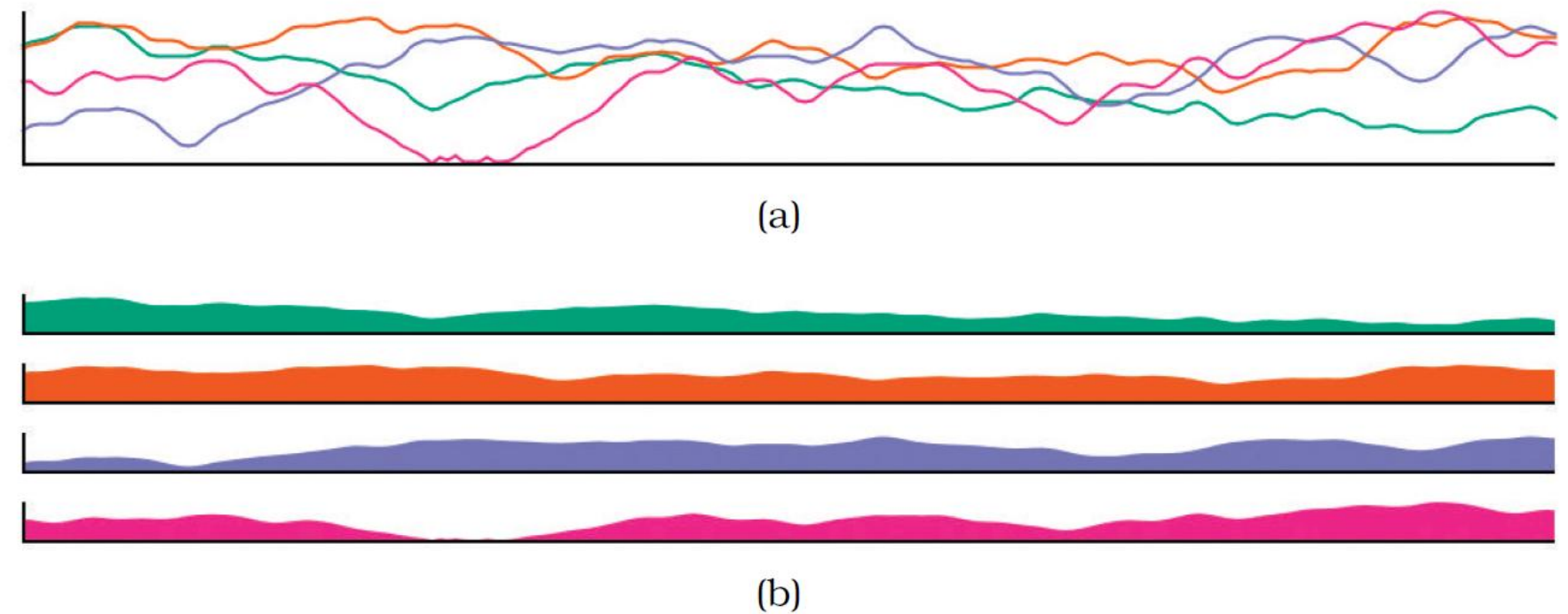
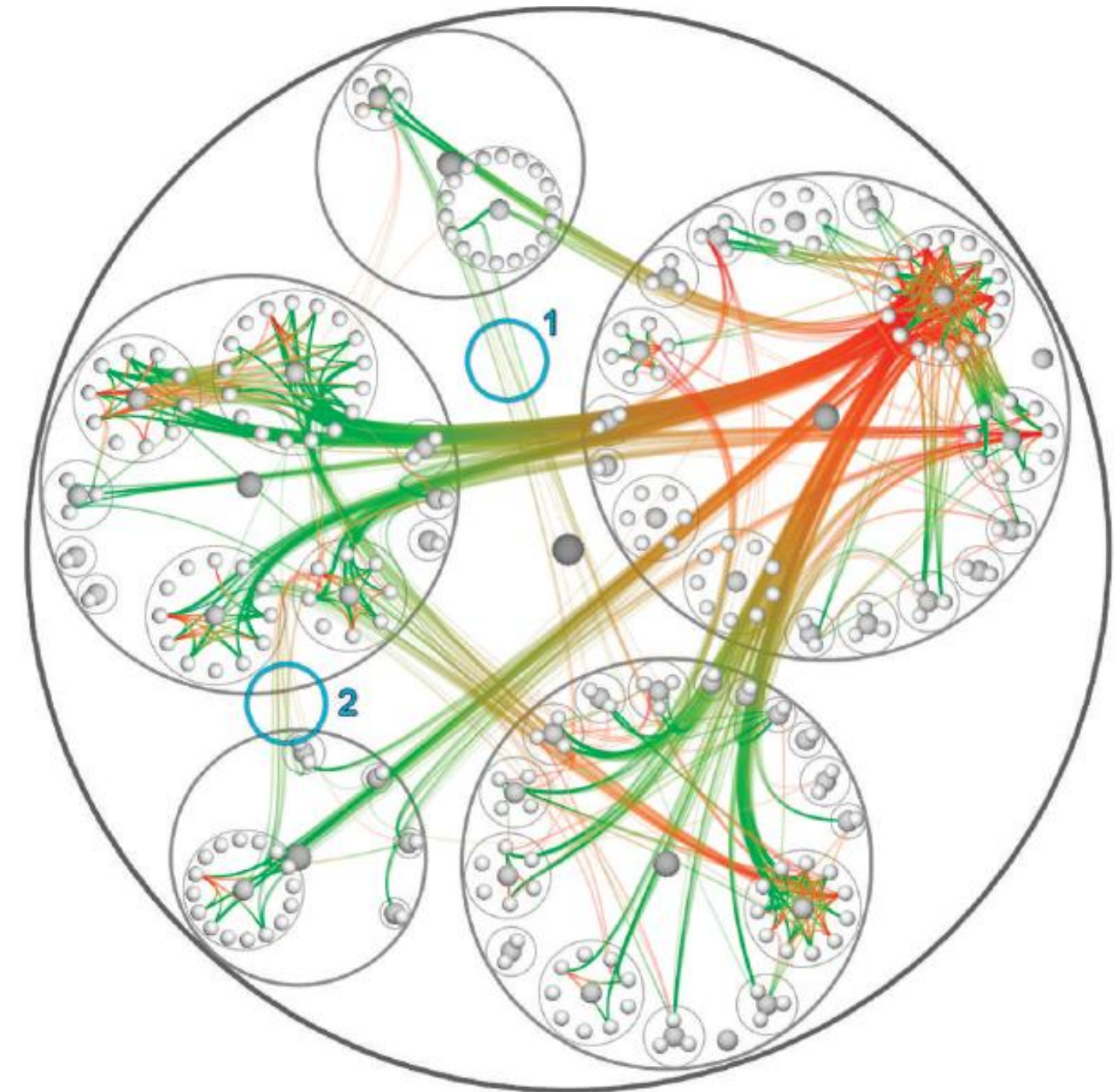


Figure 12.15: Empirical study comparing superimposed line charts to juxtaposed filled-area line chart

## Idiom - Hierarchical Edge Bundles

- The call graph network, namely, which functions call what other functions in a software system, in conjunction with the hierarchical structure of the source code
- A compound network in three layers: the tree structure in back with containment circle marks, the red-green graph edges with connection marks in a middle layer, and the graph nodes in a front layer. From [Holten 06].



# Superimpose Layers

## Dynamic visual layering

- interactive, from selection
  - lightweight: click
  - very lightweight: hover

**Cerebral** dynamically creates a **foreground visual layer of all nodes** one topological hop away in the network from the node underneath the cursor. From [Barsky et al. 07]

